

ISAR imaging of target with complex motion associated with the fractional Fourier transform

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ABSTRACT

High-resolution inverse synthetic aperture radar (ISAR) imaging based on parameter estimation of polynomial phase signal is a quite significant research hotspot, in which the azimuth echo can be modeled as multi-component quadratic frequency modulation (QFM) signal after preprocessing, which leads to the time-varying Doppler frequency. In this paper, an effective parameter estimation method called the product form of symmetric correlation function based on the fractional Fourier transform (PFrSCF) is proposed. In proposed method, a novel symmetric correlation function is used to reduce phase order of QFM signal firstly. Then, the PFrSCF can estimate two parameters of QFM signal simultaneously by the fractional Fourier transform and suppress cross term by the product in fractional Fourier transform domain. Compared with other methods, the PFrSCF is capable of suppressing cross term effectively and ensuring the good accuracy of parameters. Moreover, the PFrSCF is robust in noisy environment. Finally, associated with range-instantaneous-Doppler imaging technology, a novel ISAR imaging algorithm is presented based on PFrSCF method. The performances of PFrSCF method and the corresponding ISAR imaging algorithm of target are verified by simulated and real data.

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1. Introduction

High-resolution inverse synthetic aperture radar (ISAR) image can provide more valuable information of target in all-weather conditions, which plays a significant role in military and civil fields. It has received much attention of many researchers in the past decades [1–4]. High-quality images require fine range resolution determined by the bandwidth of transmitting signal and fine azimuth resolution that depends on the relative rotation between radar and target [5,6]. For the target with smooth motion, a well-focused ISAR image is obtained by the conventional range-Doppler (RD) algorithm [7] after range compression and motion compensation [8–10]. The conventional RD algorithm is able to reconstruct a high-quality ISAR image via the fast Fourier transform (FFT) only when Doppler frequency is a constant during the observation time.

However, in more scenes, Doppler frequency of the echo signal from the target with complex motion is time-varying, which de-

fects ISAR image by the conventional RD imaging algorithm. To improve the resolution of image, range-instantaneous-Doppler (RID) imaging technology [2,11,12] is presented and widely used in ISAR imaging system. It can be divided into two categories according to the processing technique of echo signal. The first one is known as non-parameter RID imaging technology, which can obtain ISAR image via time-frequency representation (TFR) [13–15] including short-time Fourier transform (STFT), wavelet transform (WT) and Wigner–Ville distribution (WVD) and so on. Nevertheless, these TFRs for echo signal bring a conflict between the resolution and cross-term suppression, which deteriorates the quality of ISAR image. The second category called parametric RID technique does not suffer from such problem brought by TFRs, in which azimuth echo signal can be modeled as appropriate polynomial phase signal (PPS) with phase parameters. In early days, azimuth echo of target is modeled as the linear frequency-modulated (LFM) signal, of which parameters can be estimated by several successful algorithm [16–23]. Recently, many researchers tend to model echo signal as the quadratic frequency-modulated (QFM) signal. So far several successful ISAR imaging algorithms based on parameter estimation of QFM signal [6,24–31] have been presented and generally listed into two categories: non-correlation algorithms and correla-

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tion algorithms. For non-correlation algorithms such as the maximum likelihood method in [24] and the modified discrete chirp Fourier transform in [25], they require three-dimensional searching, which leads to high computational cost inevitably. In addition, several representative correlation algorithms have been proposed by researchers. The product high-order matched-phase transform (PHMT) [26] is able to estimate parameters at lower computation cost compared with non-correlation algorithms and suppress cross term via product in transform domain. Subsequently, some improved correlation methods have been proposed such as modified cubic phase function (MCPF) in [27]. After that, other researchers interested in parametric ISAR imaging have also proposed some methods such as [28–30]. The generalized scaled Fourier transform (GSCFT) in [28] employs the nonuniform fast Fourier transform (NUFFT) along lag-time axis to suppress cross term. The modified Lv's distribution (MLVD) [29] has been proposed to estimate parameters of QFM signal and cross term has been suppressed by utilizing Keystone transform and FFT. Moreover, coherently integrated generalized cubic phase function (CIGCPF) and coherently integrated cubic phase function (CICPF) in [30] need to design an appropriate phase-terms function to improve the ability to suppress the cross term. Compared with other methods, product or integration employed in [26,27] is a simple method to suppress cross term. However, there might be error propagation because the parameters of QFM signal are estimated one by one in [26,27].

In order to solve above problems and further estimate parameters of QFM signal accurately, we propose a novel parameter estimation method called the product form of symmetric correlation function based on the fractional Fourier transform (PFRSCF). In our method, QFM signal is reduced to LFM signal without coupling between time and lag-time via symmetric correlation function (SCF). Associated with the fractional Fourier transform (FrFT) [32–36], a powerful tool in processing LFM signal, two parameters can be estimated simultaneously. Furthermore, we employ the product operation in fractional Fourier transform domain (FrFD) to suppress cross terms effectively. Based on PFRSCF method, we present a novel ISAR imaging algorithm subsequently. Compared with other method in [26,27,29], experiments with simulated data and real data validate its superior performance.

This paper is organized as follows. In Section 2, the model of echo signal is discussed for the target with complex motion. Section 3 presents the PFRSCF method in detail. In Section 4, the performance of PFRSCF is analyzed. The flowchart of ISAR imaging algorithm based on proposed PFRSCF is given and its performance is verified by simulated data and real data of target in Section 5. Section 6 is conclusion.

2. Signal model

Generally, ISAR imaging model is shown in Fig. 1 [6,26], where the Cartesian coordinates is built with X , Y , Z axes overlapped on the target and origin O is located at the rotation center of the target. R is the unit vector of the radar line of sight (LOS). ω is the 3-D rotating synthetic vector of ω_{roll} , ω_{pitch} and ω_{yaw} , which represent the angular rotation vectors of the target rotating around the X , Y , Z axes, respectively. In order to analyze the influence of rotating motion clearly, the vector ω can be decomposed into ω_e and ω_R , where ω_e and ω_R are parallel and perpendicular to R , respectively. In ISAR system, vector ω_R does not cause radial motion, and thus has no effect on Doppler frequency of echo signal. Vector ω_e is called as the effective rotation vector due to significant influence on Doppler frequency.

In Fig. 1, the scatterer P of target is denoted by vector $r_p(x, y, z)$. Then, the effective radial component caused by ω_e is

$$V_e = (\omega_e \times r_p) \cdot R \quad (1)$$

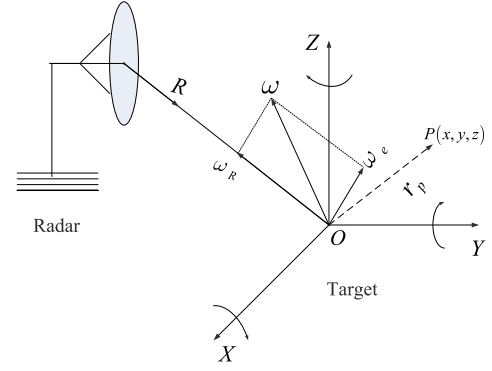


Fig. 1. ISAR imaging geometry of target with complex motion.

where “ $\times$ ” is the outer product, “ $\cdot$ ” is the inner product. Therefore, the Doppler frequency corresponding to scatterer P is

$$f_d = \frac{2V_e}{\lambda} = \frac{2(\omega_e \times r_p) \cdot R}{\lambda} \quad (2)$$

where λ denotes the wavelength of transmitted signal from radar system. ω_e is usually time-varying for the target with complex motion and can be approximated as the second-order polynomial

$$\omega_e(t) = \omega_0 + \omega_1 t + \frac{1}{2} \omega_2 t^2 \quad (3)$$

where ω_0 is the constant term, ω_1 , ω_2 are the coefficients of first-order term and second-order term, respectively, and $t \in [-T_a/2, T_a/2]$ is slow time, T_a is observation time. By substituting (3) to (1) and (2), we can obtain the time-varying velocity and Doppler frequency as follows

$$V_e(t) = (\omega_0 \times r_p) \cdot R + [(\omega_1 \times r_p) \cdot R]t + \frac{1}{2}[(\omega_2 \times r_p) \cdot R]t^2 \quad (4)$$

$$f_d(t) = \frac{2}{\lambda} \left\{ (\omega_0 \times r_p) \cdot R + [(\omega_1 \times r_p) \cdot R]t + \frac{1}{2}[(\omega_2 \times r_p) \cdot R]t^2 \right\} \quad (5)$$

According to Fig. 1 and (4), the distance from radar to the scatterer P is expressed as

$$\begin{aligned} R(t) &= \int_{t_0}^t -\frac{\lambda}{2} f_d(t) dt \\ &= \int_{t_0}^t -\left\{ (\omega_0 \times r_p) \cdot R + [(\omega_1 \times r_p) \cdot R]t \right. \\ &\quad \left. + \frac{1}{2}[(\omega_2 \times r_p) \cdot R]t^2 \right\} dt \\ &= R_0 - [(\omega_0 \times r_p) \cdot R]t - \frac{1}{2}[(\omega_1 \times r_p) \cdot R]t^2 \\ &\quad - \frac{1}{6}[(\omega_2 \times r_p) \cdot R]t^3 \end{aligned} \quad (6)$$

where t_0 denotes the initial time, R_0 denotes the initial distance from radar to the target center. After range compression and motion compensation, the azimuth echo of the scatterer P is expressed as follows

$$\begin{aligned} s_p(t) &= A_p \exp \left[-j \frac{4\pi}{\lambda} R(t) \right] \\ &= A_p \exp \{ j2\pi (a_0 + a_1 t + a_2 t^2 + a_3 t^3) \}, \end{aligned} \quad (7)$$

$$t \in [-T_a/2, T_a/2]$$

where A_p is the amplitude, $a_0 = 2 \cdot R_0/\lambda$, $a_1 = 2(\omega_0 \times r_p) \cdot R/\lambda$, $a_2 = (\omega_1 \times r_p) \cdot R/\lambda$, and $a_3 = (\omega_2 \times r_p) \cdot R/3\lambda$ are the initial phase, center frequency, chirp rate, and rate of chirp rate, respectively.

Thus, in a range cell, we can obtain the mathematical model of echo signal from all scatterers, namely multi-component QFM signal as follows

$$s(t) = \sum_{i=1}^K A_i \exp\{j2\pi(a_{i1}t + a_{i2}t^2 + a_{i3}t^3)\}, \quad (8)$$

$$t \in [-T_a/2, T_a/2]$$

where A_i is amplitude of the i th echo signal, a_{i1} , a_{i2} , a_{i3} are the phase coefficients to be estimated corresponding to phase order. In the practical situation, K is unknown and needs to be measured. Generally, one employs the residual energy error in a range bin to determine the value of K . Note that the constant phase term is omitted in (8), because it does not affect the imaging result.

3. The proposed method

In this Section, we will introduce the principle of the proposed method in detail. Firstly, the processing procedure of mono-component QFM signal and simulated results are shown in Subsection 3.1. Then as for multi-component QFM signal, the discussion and simulation are given in Subsection 3.2.

3.1. FrSCF for mono-component QFM signal

Consider the mono-component QFM signal

$$s(t) = A \exp\{j2\pi(a_1t + a_2t^2 + a_3t^3)\}, \quad t \in [-T_a/2, T_a/2] \quad (9)$$

where A denotes the amplitude of $s(t)$, a_1 , a_2 , a_3 are the coefficients to be estimated and T_a is the azimuth imaging time, namely the observation time of radar to interested target.

We employ a symmetric correlation function $SCF(t, \tau)$ of the signal $s(t)$

$$\begin{aligned} SCF(t, \tau) &= s(t + \tau)s^*(t - \tau) \cdot [s(t + \tau + \tau_0)s^*(t - \tau - \tau_0)]^* \\ &= A^4 \exp\{-j2\pi(2a_1\tau_0 + 2a_3\tau_0^3)\} \\ &\quad \cdot \exp\{-j2\pi(6a_3\tau_0^2\tau + 6a_3\tau_0\tau^2)\} \\ &\quad \cdot \exp\{-j2\pi(4a_2\tau_0t + 6a_3\tau_0t^2)\} \\ &= A^4 \phi(\tau) \cdot \exp\{-j2\pi(4a_2\tau_0t + 6a_3\tau_0t^2)\} \end{aligned} \quad (10)$$

where $\phi(\tau) = \exp[-j(2a_1\tau_0 + 2a_3\tau_0^3)] \cdot \exp[-j(6a_3\tau_0^2\tau + 6a_3\tau_0\tau^2)]$ and τ_0 is fixed non-zero lag-time difference, which is a constant within limit scope that will be discussed in the Appendix A. In (10), the first exponential term is a constant with given τ_0 and the remaining exponential terms are LFM signals about τ and t with different center frequencies and chirp rates, respectively.

Remark 1. The maximal range of τ is computed as follows

$$\begin{cases} -T_a/2 \leq t + \tau \leq T_a/2 \\ -T_a/2 \leq t - \tau \leq T_a/2 \\ -T_a/2 \leq t + \tau + \tau_0 \leq T_a/2 \\ -T_a/2 \leq t - \tau - \tau_0 \leq T_a/2 \end{cases} \implies -T_a/2 \leq \tau \leq T_a/2 - \tau_0 \quad (11)$$

Remark 2. Compared with the algorithms with coupling [6,28], there is no coupling between t and τ , which is more convenient and helpful to estimate parameters without decoupling procedure.

The FrFT of $SCF(t, \tau)$ along with t axis is defined as

$$\begin{aligned} FrSCF(u, \tau) &= \int_{-T_a/2}^{T_a/2} SCF(t, \tau) K(u, t) dt \\ &= A^4 \phi(\tau) \cdot \int_{-T_a/2}^{T_a/2} \exp\{-j2\pi(4a_2\tau_0t + 6a_3\tau_0t^2)\} \\ &\quad \cdot K(u, t) dt \\ &= A^4 A_\alpha \phi(\tau) \exp(j\pi u^2 \cot \alpha) \\ &\quad \cdot \int_{-T_a/2}^{T_a/2} \exp\{j\pi(\cot \alpha - 12a_3\tau_0)t^2\} \\ &\quad \cdot \exp\{-j2\pi(u \csc \alpha + 4a_2\tau_0)t\} dt \\ &\stackrel{\cot \alpha = 12a_3\tau_0}{=} T_a A_\alpha A^4 \phi(\tau) \exp(j\pi u^2 \cot \alpha) \\ &\quad \cdot \text{sinc}[\pi T_a(u \csc \alpha + 4a_2\tau_0)]. \end{aligned} \quad (12)$$

In (12), $K(u, t)$ denotes the kernel function of the FrFT

$$\begin{aligned} K_\alpha(u, t) &= \begin{cases} A_\alpha \exp\{j\pi(t^2 + u^2) \cot \alpha - j2\pi t u \csc \alpha\}, & \alpha \neq k\pi \\ \delta(t - u), & \alpha = 2k\pi \\ \delta(t + u), & \alpha = (2k - 1)\pi, \end{cases} \end{aligned} \quad (13)$$

and $A_\alpha = \sqrt{1 - j \cot \alpha}$. The rotation angle of FrFT is expressed as $\alpha = p \cdot \pi/2$ and p is the order of FrFT. From (12), $FrSCF(u, \tau)$ with the matched angle α is capable of gathering the energy to be a straight line paralleling to τ . Because there is no coupling between t and τ luckily, our method can find the position of peak at arbitrary τ easily to estimate the parameters.

According to (12), the energy of signal will be gathered when $\cot \alpha = 12a_3\tau_0$. So we can obtain

$$\begin{cases} \cot \alpha = 12a_3\tau_0 \\ u \csc \alpha = -4a_2\tau_0 \end{cases} \quad (14)$$

From (14), therefore, the position of peak is determined by α and u . We can estimate two parameters simultaneously by the FrSCF method as follows

$$\begin{cases} \hat{a}_3 = \frac{\cot \alpha}{12\tau_0} \\ \hat{a}_2 = -\frac{u \csc \alpha}{4\tau_0}. \end{cases} \quad (15)$$

After obtaining the parameters a_2 and a_3 by the proposed method, the remaining parameters in the corresponding QFM signal can be easily estimated by the dechirp technique and the Fourier transform (FT) when compensating for the chirp term and chirp rate term [6,26,27].

It is imperative to demonstrate the performance of the proposed method by simulated results at first. If sample interval is Δt , the free-noise discrete signal is

$$s(n\Delta t) = A \exp\{j2\pi[a_1n\Delta t + a_2(n\Delta t)^2 + a_3(n\Delta t)^3]\} \quad (16)$$

where N is the sample number and is assumed to be odd e.g. $N = 513$, and then $n = -(N - 1)/2, \dots, (N - 1)/2$. Naturally, $SCF(n\Delta t, m\Delta t)$ and $FrSCF(n\Delta u, m\Delta t)$ correspond to $SCF(t, \tau)$ and $FrSCF(u, \tau)$, respectively. Note that the maximal range of $m = -(N - 1)/2, \dots, (N - 1)/2 - \tau_0$.

Table 1

The parameters of simulated signal.

Parameter	A	a_1	a_2	a_3
Value	1	1/64	1/5N	1/10N ²

Remark 3. To avoid the ambiguity brought by the periodicity of spectra of sampled signals, it is necessary to discuss the range of parameters of QFM signal [6,38]. We have

$$|a_1| \leq \frac{1}{2\Delta t} \quad (17)$$

$$|a_2| \leq \frac{1}{2N(\Delta t)^2} \quad (18)$$

$$|a_3| \leq \frac{2}{3N^2(\Delta t)^3}. \quad (19)$$

Assuming that normalized sample interval $\Delta t = 1$, the parameters of QFM signal are shown in Table 1 and Fig. 2 gives the simulation results. Fig. 2(a) is the distribution with respect to $SCF(t, \tau)$ in t - τ plane. Fig. 2(b) is $FrSCF(u, \tau)$ in u - τ plane with the matched angle, which is a straight line paralleling to τ . As we have already analyzed theoretically, different τ do not influence on gathering energy of signal. Assumed that $\tau = -\tau_0/2$ here, we can obtain 3-D $FrSCF(u, \tau)$ and 2-D $FrSCF(u, \tau)$ of $s(t)$ shown in Fig. 2(c) and Fig. 2(d) respectively, from which the parameters can be estimated well by obtaining the location of peak.

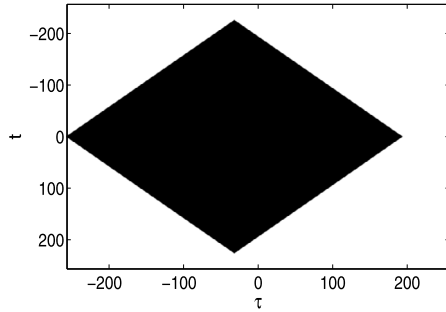
3.2. PFrSCF for multi-component QFM signals

Without loss of the generality, we discuss the two-component QFM signal

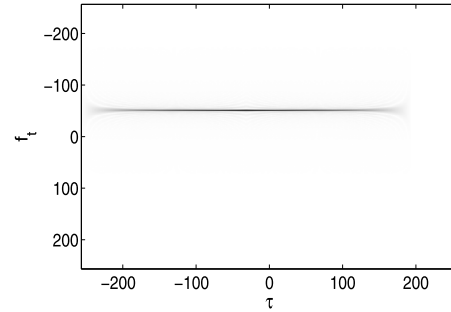
$$s(t) = \sum_{i=1}^2 A_i \exp\{j2\pi(a_{i1}t + a_{i2}t^2 + a_{i3}t^3)\} \quad (20)$$

Thus, the SCF of signal in (20) is written as follows

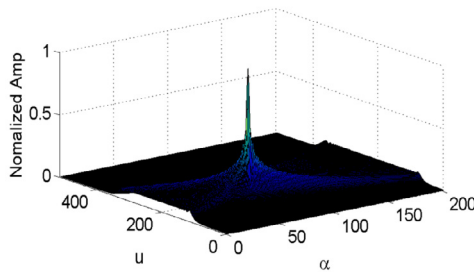
$$SCF(t, \tau) = SCF_{auto}(t, \tau) + SCF_{cross}(t, \tau) \quad (21)$$



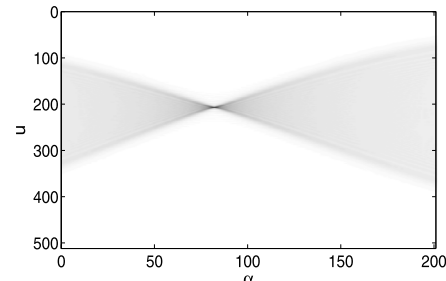
(a)



(b)



(c)



(d)

Fig. 2. The results by the $FrSCF$. (a) the SCF result in τ - t plane, (b) the $FrSCF$ result with matched order, (c) the 3-D $FrSCF$ result, (d) the grayscalemap of $FrSCF(u, \tau)$.

where the auto term is

$$\begin{aligned} SCF_{auto}(t, \tau) &= \sum_{i=1}^2 A_i^4 \exp\{-j2\pi(2a_{i1}\tau_0 + 2a_{i3}\tau_0^3)\} \\ &\quad \cdot \exp\{-j2\pi(6a_{i3}\tau_0^2\tau + 6a_{i3}\tau_0\tau^2)\} \\ &\quad \cdot \exp\{-j2\pi(4a_{i2}\tau_0t + 6a_{i3}\tau_0t^2)\} \\ &= \sum_{i=1}^2 A_i^4 \phi_i(\tau) \cdot \exp\{-j2\pi(4a_{i2}\tau_0t + 6a_{i3}\tau_0t^2)\} \end{aligned} \quad (22)$$

of which $\phi_i(\tau) = \exp\{-j2\pi(2a_{i1}\tau_0 + 2a_{i3}\tau_0^3)\} \exp\{-j2\pi(6a_{i3}\tau_0^2\tau + 6a_{i3}\tau_0\tau^2)\}$. $SCF_{cross}(t, \tau)$ denotes the cross term, which will be discussed in Section 4 and Appendix B. The FrFT of $SCF(t, \tau)$ along with t axis is shown as follows

$$\begin{aligned} FrSCF(u, \tau) &= \int_{-T_a/2}^{T_a/2} SCF(t, \tau) K(u, t) dt \\ &= FrSCF_{auto}(u, \tau) + FrSCF_{cross}(u, \tau) \end{aligned} \quad (23)$$

where

$$\begin{aligned} FrSCF_{auto}(u, \tau) &= \int_{-T_a/2}^{T_a/2} SCF_{auto}(t, \tau) K(u, t) dt \\ &= \sum_{i=1}^2 T_a A_i^4 A_{\alpha_i} \phi_i(\tau) \exp\{j\pi u^2 \cot \alpha_i\} \\ &\quad \cdot \text{sinc}[\pi T_a(u \csc \alpha_i + 2a_{i2}\tau_0)]. \end{aligned} \quad (24)$$

$FrSCF_{auto}(u, \tau)$ and $FrSCF_{cross}(u, \tau)$ denote the auto terms and cross terms generated by (21) via FrFT.

Table 2

The parameters of two simulated signal.

Parameter	A_i	a_{i1}	a_{i2}	a_{i3}
$s_1(t)$	1	1/64	1/5N	1/2N ²
$s_2(t)$	1	-1/64	-3/20N	1/5N ²

Similar to (12), it is easy to find that the energy of auto terms in (24) can be gathered to be peaks independent of τ . However, cross terms do not share this property and the specific analysis can be referred to Section 4 and Appendix B. Hence, in order to suppress the cross term of multi-component QFM signal, we can product $FrSCF(u, \tau)$ in FrFD while choosing different τ , which can enhance peaks of auto term, i.e.,

$$PFrSCF(u) = \prod_{l=1}^L FrSCF(u, \tau_l) \quad (25)$$

where L is the number of chosen τ .

For a free-noise discrete two-component QFM signal with the length $N = 513$

$$s(n\Delta t) = \sum_{i=1}^2 A_i \exp\{j2\pi[a_{i1}n\Delta t + a_{i2}(n\Delta t)^2 + a_{i3}(n\Delta t)^3]\} \quad (26)$$

Assuming the sample interval $\Delta t = 1$ and $n = -(N-1)/2, \dots, (N-1)/2$, the parameters in (26) are shown in Table 2. As shown in Fig. 3, the aggressive cross terms occur in the results of FrFT. The results of $PFrSCF$ are shown in Fig. 4. Fig. 4(a) is the 3-D result processed by $PFrSCF$ method, which suppress cross terms obviously. And grayscalemap is shown in Fig. 4(b). It is important to notice the fact that the product in FrFD has considerably strong ability of suppressing cross terms, which is beneficial to ISAR imaging of target with the complex motion.

In order to suppress the cross term effectively, L and τ in the proposed method should be selected appropriately. It is the

suppression ability and the consumed time that are mainly considered. Hence, we first discuss the selection of L . If L is too small, it is so difficult to suppress the cross term effectively, and if too large, it leads to increase of computational cost. As for the selection of τ , we mainly discuss the interval of adjacent τ . If the step size of τ is large relatively, it limits the selection of L , and L has to be small. Nevertheless, if the step size of τ is small, energy distributions of cross terms for different τ are similar, so the energy of cross terms will be gathered when using product operator. In a word, for the sake of suppressing cross term and computational cost, L and τ should be selected according to practical requirement. In Fig. 4, we select $L = 15$ and the interval of τ is 15.

Moreover, in fact, it is $\cot\alpha$ that directly affect the FrFT rather than α . We utilize uniform $\cot\alpha$ in Fig. 5, where the number of product to suppress cross terms is decreased and $L = 12$ can obtain good performance.

4. Performance analysis of the proposed method

4.1. Suppression of cross term

For the two-component QFM signal in (20), we can divide cross term in (21) and (23) into four categories based on the combination of signals. Therefore, the cross terms can be shown as follows

$$SCF_{cross}(t, \tau) = SCF_{1cross}(t, \tau) + SCF_{2cross}(t, \tau) + SCF_{3cross}(t, \tau) + SCF_{4cross}(t, \tau) \quad (27)$$

and

$$FrSCF_{cross}(t, \tau) = FrSCF_{1cross}(t, \tau) + FrSCF_{2cross}(t, \tau) + FrSCF_{3cross}(t, \tau) + FrSCF_{4cross}(t, \tau). \quad (28)$$

In order to validate the effect of the proposed method on suppressing cross term, we might as well take the first category

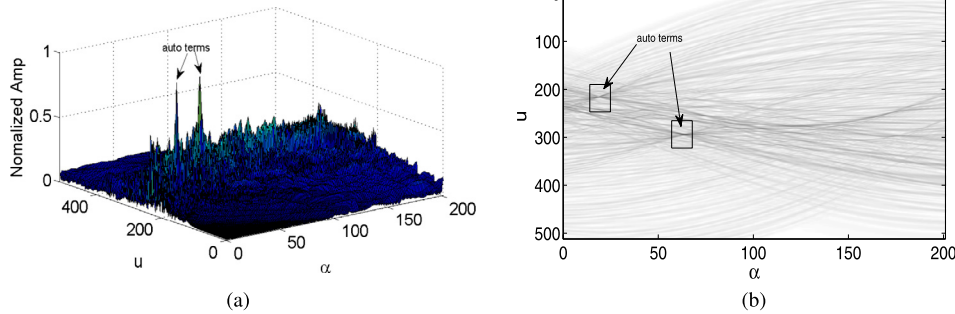


Fig. 3. The results by the FrSCF: without product. (a) the 3-D result, (b) the grayscalemap in u - α plane.

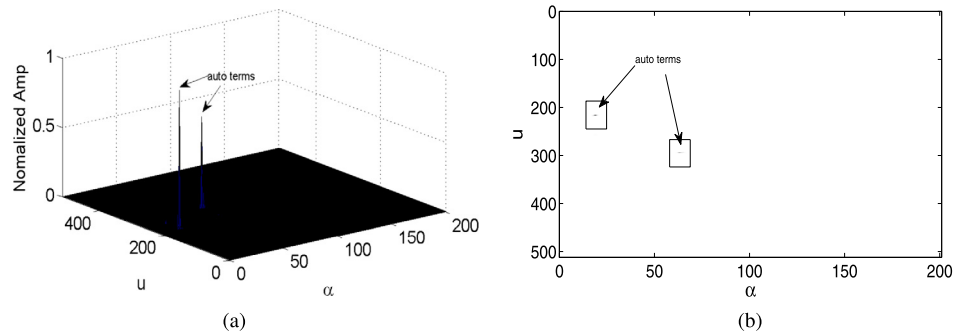


Fig. 4. The results by the PFrSCF. (a) the 3-D result, (b) the grayscalemap in the u - α plane.

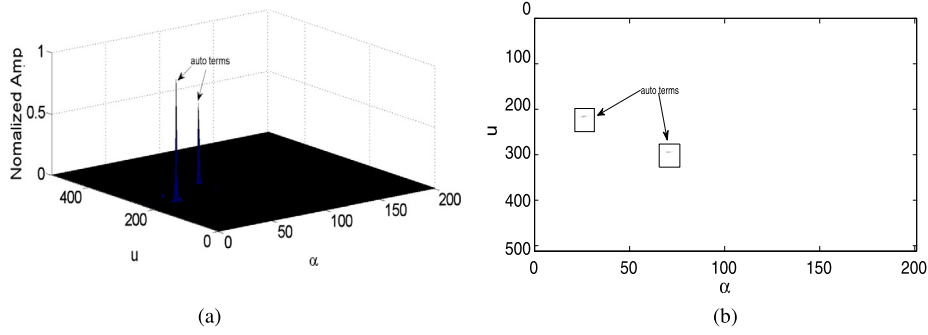


Fig. 5. The results by the PFrSCF with non-uniform α . (a) the 3-D result, (b) the grayscale map in the u - α plane.

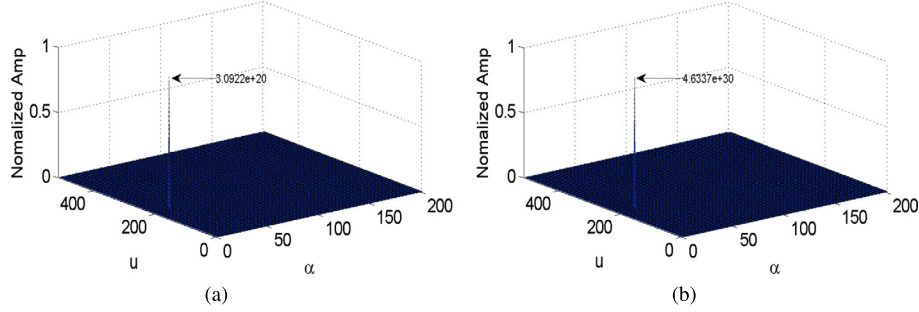


Fig. 6. The results via PFrSCF. (a) only s_1 , (b) when $a_{12} = a_{22}$ and $a_{13} = a_{23}$.

$SCF_{1cross}(t, \tau)$ for example to analyze here, which help us distinguish the true peaks and spurious peaks. As for the others, readers can refer to Appendix B.

The first category of cross terms is

$$SCF_{1cross}(t, \tau) = \sum_{i=1}^2 \sum_{\substack{j=1 \\ i \neq j}}^2 |A_i^2| |A_j^2| \cdot \exp[-j2\pi(2a_{j1}\tau_0 + 2a_{j3}\tau_0^3)] \\ \cdot \exp\{-j2\pi[(2a_{j1} - 2a_{i1} + 6a_{j3}\tau_0^2)\tau \\ + 6a_{j3}\tau_0\tau^2 + 2(a_{i3} - a_{j3})\tau^3]\} \\ \cdot \exp\{j2\pi\{[(4(a_{i2} - a_{j2})\tau - 4a_{j2}\tau_0)]t \\ + [6(a_{i3} - a_{j3})\tau - 6a_{j3}\tau_0]t^2]\} \} \quad (29)$$

Applying FrFT to (29) along t axis, we can obtain $FrSCF_{1cross}(u, \tau)$ and the positions of peaks as follows

$$\begin{cases} \cot \alpha = -12[(a_{i3} - a_{j3})\tau - a_{j3}\tau_0] \\ u \csc \alpha = 4[(a_{i2} - a_{j2})\tau - a_{j2}\tau_0] \end{cases} \quad (30)$$

As analyzed in Section 3, the peak position of auto term is independent of τ . From (30), however, it is easy to find that the peak positions of cross term are related to τ , which motivates us to suppress the cross term by the product operator in FrFD along τ . It is imperative to discuss some cases where parameters of different QFM signal may be the same in practical environment. This discussion about suppressing the cross terms is important for imaging of target further, because the cross terms correspond to artificial scatterers.

Case 1a. If $a_{i3} = a_{j3}$ and $a_{i2} = a_{j2}$, (30) can be rewritten as

$$\begin{cases} \cot \alpha = 12a_{j3}\tau_0 \\ u \csc \alpha = -4a_{j2}\tau_0 \end{cases} \quad (31)$$

which can enhance energy of the corresponding auto term. The results including only signal $s_1(1)$ and Case 1a are shown in Fig. 6.

Case 1b. If $a_{i3} \neq a_{j3}$ or $a_{i2} \neq a_{j2}$, (30) can be rewritten as

$$\begin{cases} \cot \alpha = -12[(a_{i3} - a_{j3})\tau - a_{j3}\tau_0] \\ u \csc \alpha = -4a_{j2}\tau_0 \end{cases} \quad (32)$$

or

$$\begin{cases} \cot \alpha = 12a_{j3}\tau_0 \\ u \csc \alpha = 4[(a_{i2} - a_{j2})\tau + a_{j2}\tau_0] \end{cases} \quad (33)$$

where the peak position is also associated with τ . So we employ product $FrSCF_{1cross}(u, \tau)$ at different τ to suppress the spurious peaks and enhance true peaks perfectly. The results including $a_{12} = a_{22}$ or $a_{13} = a_{23}$ are shown in Fig. 7.

Case 1c. If $a_{i3} \neq a_{j3}$ and $a_{i2} \neq a_{j2}$, (30) maintains to be the general form

$$\begin{cases} \cot \alpha = -12[(a_{i3} - a_{j3})\tau - a_{j3}\tau_0] \\ u \csc \alpha = 4[(a_{i2} - a_{j2})\tau - a_{j2}\tau_0] \end{cases} \quad (34)$$

where α or u are associated with τ in $FrSCF_{1cross}(u, \tau)$ simultaneously. In such a case, the peak of cross-terms appear the different position in u - α plane according to different τ . The simulation results can refer to Fig. 4.

4.2. Robustness in noise

In the practical environment, there are various noises inevitably between radar and target. Therefore, it is necessary to analyze the robustness of the proposed PFrSCF method in noisy environment and compare it with other methods such as PHMT method [26], MCPF method [27] and MLVD method [29] in this Subsection. Besides, the mean square error (MSE) is employed to evaluate robustness of the PFrSCF method as well as the PHMT method, MCPF method and MLVD method. The results of parameter a_3 and a_2 are shown in Fig. 8(a) and Fig. 8(b), respectively.

The simulated signal $s_1(t)$ in Table 2, embedded in white Gaussian noise, is estimated by these four methods in a wide range of

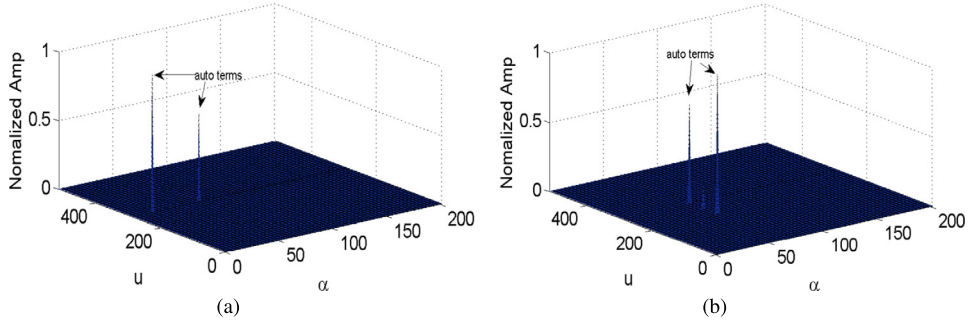


Fig. 7. The results in Case 2a. (a) when $a_{12} = a_{22}$, (b) when $a_{13} = a_{23}$.

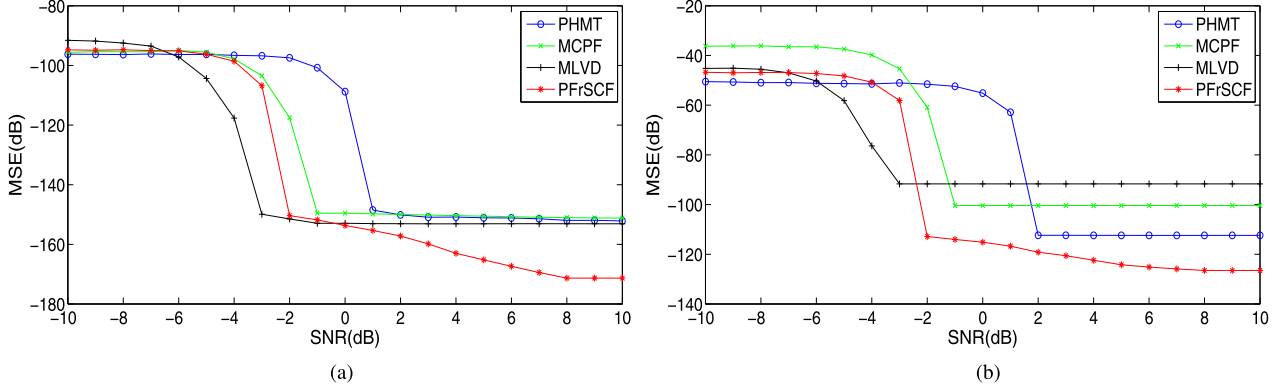


Fig. 8. Performances of different methods in noise. (a) the MSE of a_3 , (b) the MSE of a_2 .

Table 3
The computational cost of methods.

Method	PHMT	MCPF	MLVD	PFrSCF
Computational cost	$O(N^3)$	$O(N^3)$	$O(N^2 \log_2 N)$	$O(MN^2 \log_2 N)$

signal-to-noise ratio (SNR) from -10 dB to 10 dB. There are 1000 trials in Monte Carlo simulation. As shown in Fig. 8(a) and (b), the MSE decreases very rapidly at a certain SNR as these methods suffer the threshold effect [37,38]. Compared with these four methods, the proposed PFrSCF method has good performance in terms of threshold and MES of parameters, which implies that it can obtain better accuracy of parameter with wide SNR range.

4.3. Computational cost

The computational cost of our proposed method will be analyzed in this subsection. Its main implementation procedures include the symmetric correlation function ($O(N^2)$), the FrFT operation ($M \cdot O(N^2 \log_2 N)$) and product operation ($O(N^2)$), where N denotes sampling number and M is searching times. Thus, the computational cost of proposed method is $O(N^2) + M \cdot O(N^2 \log_2 N) + O(N^2) = O(MN^2 \log_2 N)$. Computational costs of PHMT method [26] and MCPF method [27] are both $O(N^3)$, which are brought by non-uniformly Fourier transform along with lag axis. In MLVD method [29], 2-D FFT operation with ($O(N^2 \log_2 N)$) is necessary. Table 3 lists the computational costs of PHMT method, MCPF method, MLVD method and proposed PFrSCF method. When $N > M \log_2 N$, the proposed PFrSCF method has lower computational cost than PHMT method and MCPF method.

5. ISAR imaging algorithm

In this Section, we induce the detailed procedure of ISAR imaging algorithm based on proposed PFrSCF method and verify its effectiveness via simulated data and real data.

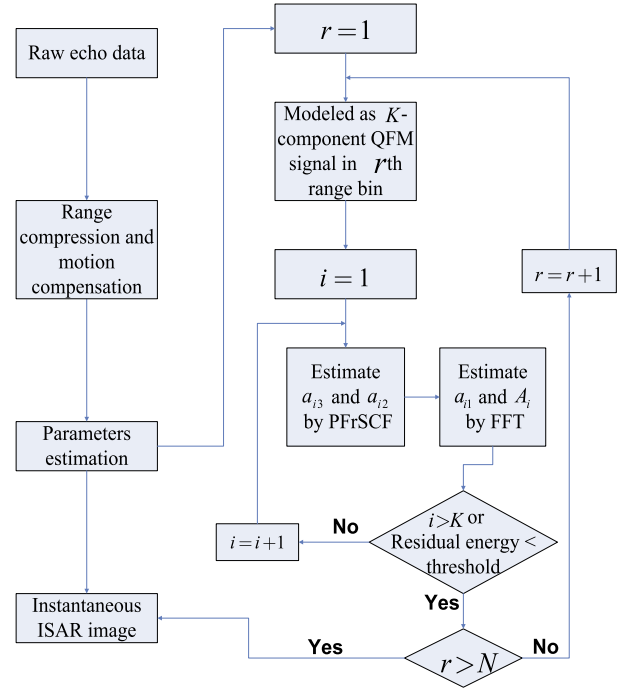


Fig. 9. The flowchart of ISAR imaging algorithm based on PFrSCF method.

5.1. Implementation of ISAR imaging algorithm

The flowchart of the ISAR imaging algorithm based on the PFrSCF method is shown in Fig. 9. The implementation procedure corresponding to the flowchart is illustrated in detail as follows.

Step. 1: Perform the range compression and the motion compensation to the radar echo data.

Step. 2: Assume that the received echo signal in r th range bin can be regarded as

$$s(t) = \sum_{i=1}^K A_i \exp[j2\pi(a_{i1}t + a_{i2}t^2 + a_{i3}t^3)] \quad (35)$$

where K is the component number of the QFM signal in the r th range bin, A_i denotes the amplitude and a_{i1} , a_{i2} , a_{i3} are the coefficients of the i th component, respectively.

Step. 3: Initialize $i = 1$,

$$s_i(t) = A_i \exp[j2\pi(a_{i1}t + a_{i2}t^2 + a_{i3}t^3)] \quad (36)$$

Step. 4: Apply the PFrSCF method to $s_i(t)$, then estimate the parameters and obtain $\hat{a}_{i3}$ and $\hat{a}_{i2}$ simultaneously.

Step. 5: Construct the compensation function with high-order phase for $s_i(t)$ in Step. 2

$$s_{com}(t) = \exp[-j2\pi(\hat{a}_{i2}t^2 + \hat{a}_{i3}t^3)] \quad (37)$$

and obtain the sinusoid signal with parameter a_{i1} by

$$s_{sin} = s(t) \cdot s_{com}(t). \quad (38)$$

Step. 6: Apply the FFT to s_{sin} to obtain the estimation of a_{i1} .

Step. 7: Estimate the amplitude A_i by

$$\hat{A}_i = \frac{1}{N} \left| \sum_{-N/2}^{N/2-1} s_i(t) \cdot \exp[-j2\pi(\hat{a}_{i1}t + \hat{a}_{i2}t^2 + \hat{a}_{i3}t^3)] \right| \quad (39)$$

where $t = n \cdot \Delta t$, $n = -N/2, \dots, N/2 - 1$ and Δt is the sample interval.

Step. 8: Reconstruct the i th component of QFM signal with the estimated parameters

$$s_i(\hat{t}) = \hat{A}_i \cdot \exp[-j2\pi(\hat{a}_{i1}t + \hat{a}_{i2}t^2 + \hat{a}_{i3}t^3)], \quad (40)$$

and then subtract it from $s(t)$ by the notch filter with proper center and bandwidth.

Step. 9: Let $i = i + 1$ and repeat Step. 3–8 until $i = K$. If the K is unknown, repeat Step. 3–8, and then end up until that the residual energy is less than a fixed threshold in the r th range bin.

Step. 10: Let $r = r + 1$ and repeat Step. 2–9 until $r = N$.

Step. 11: The well-focused ISAR image can be obtained.

Table 4

The parameter of radar and target.

Radar	Target		
Center frequency (GHz)	10	Range (km)	10
Bandwidth (MHz)	150	Target (m)	100
PRF (Hz)	500	Velocity of rotation (rad/s)	0.01
Pulse length (μ s)	0.5	Acceleration of rotation (rad/s ²)	0.08
Range bin number	400	Acceleration rate of rotation (rad/s ³)	0.03
Pulse number	400		

5.2. ISAR imaging based on simulated data

In this Subsection, we use the simulated data to evaluate the proposed ISAR imaging algorithm. The parameters of radar and target are shown in Table 4. In order to further ensure the feasibility in practical environment, white Gaussian noise is embedded into the echo signals and SNR is -1 dB. Fig. 10(a) is a simple ship target model and the traditional RD image is shown in Fig. 10(b), which is blurred severely. As presented in Figs. 11–14, ISAR images can be reconstructed based on PHMT [26], MCPF [27], MLVD [29] and PFrSCF, respectively.

Moreover, the quality of ISAR image, generally, would be measured in a quantitative criterion like the entropy of image employed in [6,26,27,29]

$$Entropy = \sum_{m=1}^M \sum_{n=1}^N \frac{|I(m, n)|^2}{S} \ln \frac{S}{|I(m, n)|^2} \quad (41)$$

where

$$S = \sum_{m=1}^M \sum_{n=1}^N |I(m, n)|^2. \quad (42)$$

And the smaller the entropy is, the better the quality of image is. The ISAR images are measured by entropy and the results corresponding to Figs. 11–14 are shown in Table 5.

The PFrSCF method, based on four-order correlation function, is able to estimate parameters a_3 and a_2 simultaneously by FrFT and simple multiplication, which can avoid error propagation and obtain good performance in suppressing cross term. From Figs. 11–14 and Table 5, the proposed PFrSCF method produces ISAR images with better quality. Thus, the proposed ISAR imaging algorithm based on PFrSCF method is considerable to reconstruct ISAR image of high quality in practical situation.

5.3. ISAR imaging based on real data

The real data can be used to validate the performance of proposed ISAR imaging algorithm. The basic parameters of radar and target are shown in Table 6 and the results of received echo data after range compression and motion compensation are given in Fig. 15.

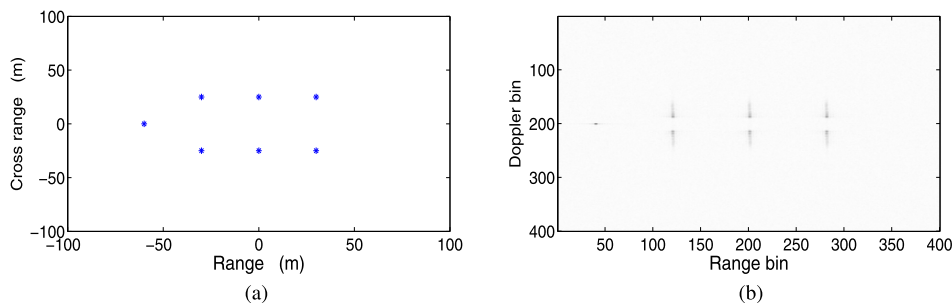


Fig. 10. ISAR images: (a) the target, (b) RD image.

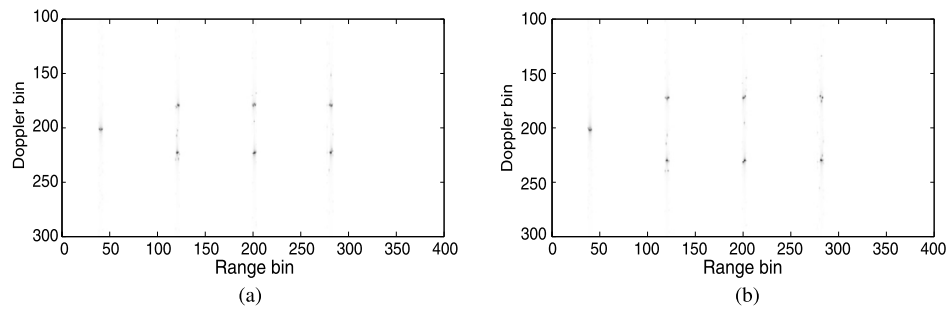


Fig. 11. ISAR images based on the PHMT algorithm: (a) $t = 0.1920$ s, (b) $t = 0.2820$ s.

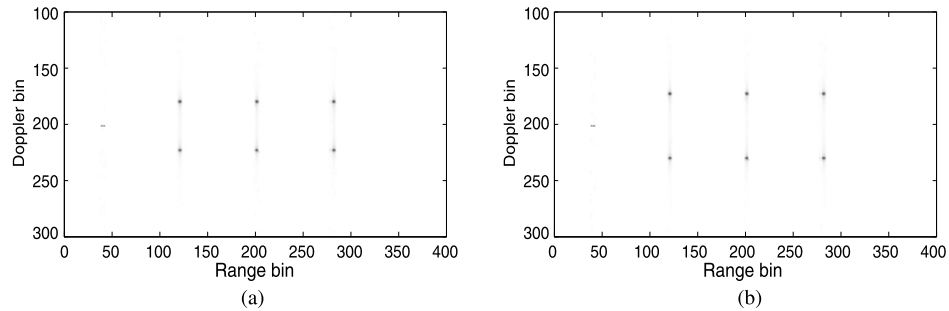


Fig. 12. ISAR images based on the MCPF algorithm: (a) $t = 0.1920$ s, (b) $t = 0.2820$ s.

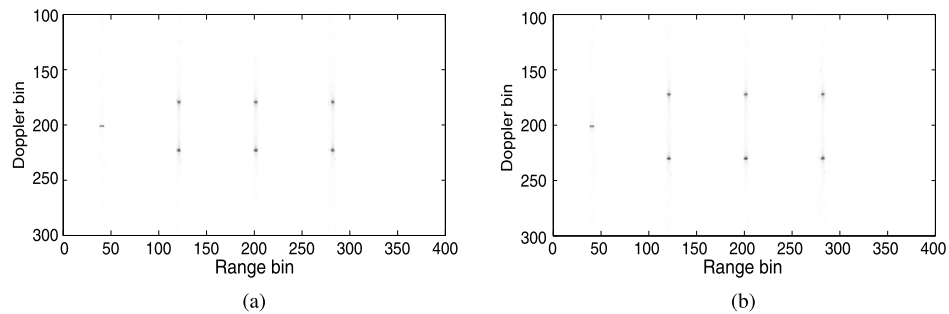


Fig. 13. ISAR images based on the MLVD algorithm: (a) $t = 0.1920$ s, (b) $t = 0.2820$ s.

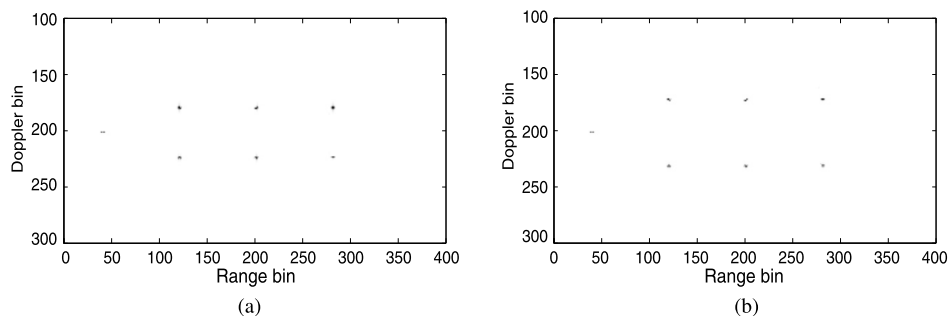


Fig. 14. ISAR images based on the PFrSCF algorithm: (a) $t = 0.1920$ s, (b) $t = 0.2820$ s.

Table 5

The entropies of images based on simulated data.

Algorithm	$t = 0.1920$ s	$t = 0.2820$ s
PHMT	4.6580	4.6946
MCPF	4.4928	4.5073
MLVD	4.4613	4.2213
PFrSCF	4.2691	3.8458

Table 6

The parameter of radar and target.

Radar		Ship target	
Center frequency (GHz)	9.25	Size of ship (m)	24
Bandwidth (MHz)	500	Velocity (m/s)	8
PRF (Hz)	200	Range (km)	6
Pulse length (μ s)	600		
Range bin number	256		
Pulse number	256		

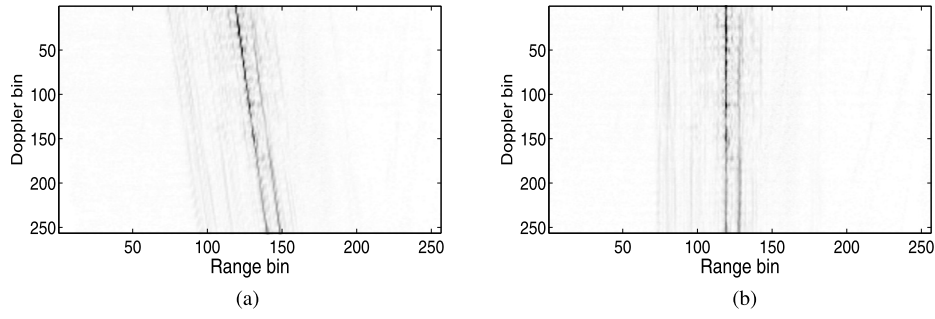


Fig. 15. Images after preprocessing. (a) range compression, (b) motion compensation.

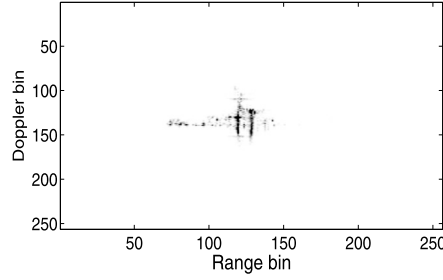


Fig. 16. RD image.

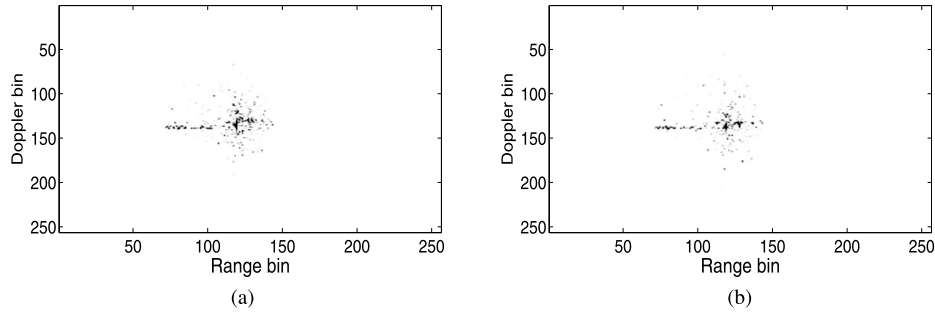


Fig. 17. ISAR images based on the PHMT algorithm. (a) $t = 0.1950$ s, (b) $t = 0.2700$ s.

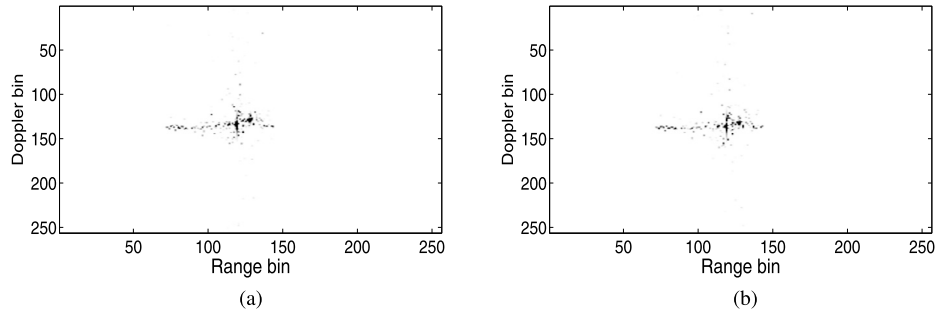


Fig. 18. ISAR images based on the MCPF algorithm. (a) $t = 0.1950$ s, (b) $t = 0.2700$ s.

Traditional RD image with much more spurious peaks, which is blurry, is obtained in Fig. 16. As presented in Figs. 17–20, the ISAR images are reconstructed by ISAR imaging algorithm based on PHMT, MCPF and PFrSCF methods, respectively. Through applying the proposed ISAR imaging algorithm based on PFrSCF method to real data from the practical marine environment, ISAR images at different time can be reconstructed, which are measured by the entropy that are shown in Table 7. According to the above results, one can find the fact that proposed ISAR imaging algorithm based on PFrSCF method has a superior performance, which can be applied to practical environment.

6. Conclusion

In this paper, we present an ISAR imaging algorithm based on the proposed PFrSCF method. For target with complex motion, echo signal in a range bin is modeled as multicomponent QFM signal after range compression and motion compensation. The proposed PFrSCF method can estimate accurately the two parameters of QFM signal simultaneously by SCF and FrFT. Subsequently, its product form in FrFD is employed to suppress the cross term effectively. Furthermore, the characters of cross term is discussed. Experiments with simulated data and real data show that the pro-

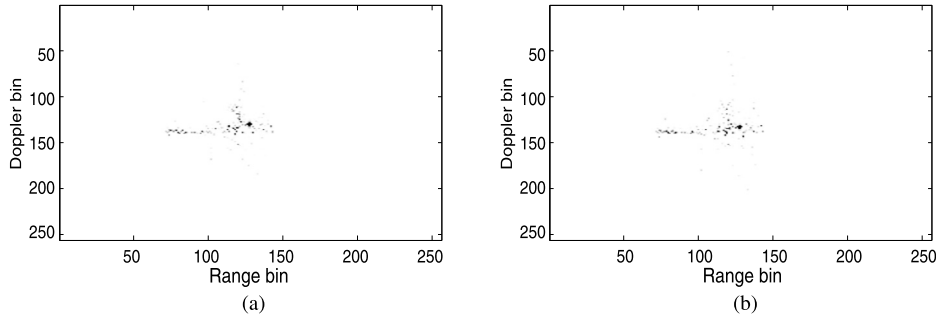


Fig. 19. ISAR images based on the MLVD algorithm. (a) $t = 0.1950$ s, (b) $t = 0.2700$ s.

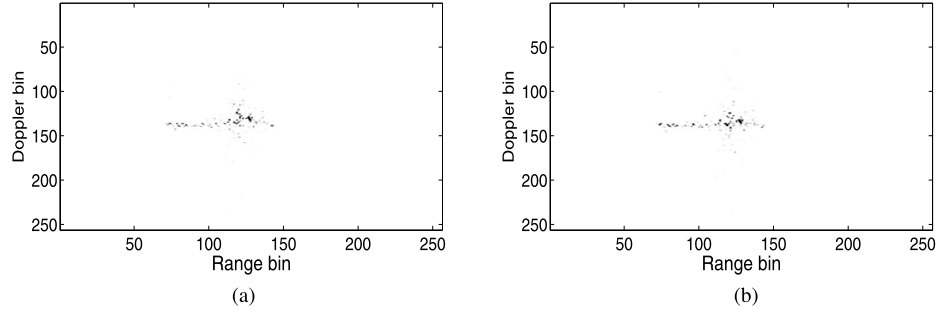


Fig. 20. ISAR images based on the PFrSCF algorithm. (a) $t = 0.1950$ s, (b) $t = 0.2700$ s.

Table 7

The entropies of images based on real data.

Algorithm	$t = 0.1950$ s	$t = 0.2700$ s
PHMT	4.5335	4.5614
MCPF	4.2001	4.1484
MLVD	4.1918	4.2209
PFrSCF	3.6527	3.6607

posed ISAR imaging algorithm based on PFrSCF has superior performance.

Acknowledgments

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Appendix A

In this appendix, we discuss the criterion for the selection of the fixed lag time τ_0 . Larger τ_0 can improve parameter estimation precision, if not, it is possible to lead to the spectrum aliasing [6, 39].

For a QFM signal

$$s(t) = A \exp[j2\pi(a_1 t + a_2 t^2 + a_3 t^3)], \quad (\text{A.1})$$

the Doppler spectrum is

$$D(t) = 2a_2 t + 3a_3 t^2. \quad (\text{A.2})$$

To avoid $D(t)$ exceeding an entire PRF, according to [6,18], we have

$$|a_2| \leq \frac{PRF}{2T_a} \quad \text{and} \quad |a_3| \leq \frac{2PRF}{3T_a^2} \quad (\text{A.3})$$

By using the proposed method, a_3 and a_2 can be estimated as

$$\begin{cases} \hat{a}_3 = \frac{\cot \alpha}{12\tau_0} \\ \hat{a}_2 = -\frac{u \csc \alpha}{4\tau_0}, \end{cases} \quad (\text{A.4})$$

where $\alpha \in [\pi/4, 3\pi/4]$, $|u \csc \alpha| \leq \frac{PRF}{2}$. Therefore,

$$\begin{cases} \max(|\hat{a}_3|) = \frac{1}{12\tau_0} \\ \max(|\hat{a}_2|) = \frac{PRF}{8\tau_0} \end{cases} \quad (\text{A.5})$$

Associated with Eq. (A.3), we have obviously

$$\begin{cases} \frac{1}{12\tau_0} \geq \frac{2PRF}{3T_a^2} \\ \frac{PRF}{8\tau_0} \geq \frac{PRF}{2T_a} \end{cases} \quad (\text{A.6})$$

Then we can obtain

$$\tau_0 \leq \frac{T_a}{4} \quad (\text{A.7})$$

For the proposed method, large τ_0 would generate the spectrum ambiguity. Selecting a proper fixed lag time τ_0 can avoid the spectrum problem and ensure the estimation accuracy as far as possible. We therefore choose $\tau_0 = T_a/8$ in this paper.

Appendix B

The first category of cross term has been analyzed in Section 4 and the FrFT is applied to $SCF(t, \tau)$ along t axis, therefore, we just need to concentrate on the exponential terms about t in the cross term. The other cross term categories can be discussed here. In order to facilitate the analysis, $\tau + \tau_0$ can be written as τ_1 in the remaining cross term.

The second category of cross term is

$$\begin{aligned}
 SCF_{2cross}(t, \tau, \tau_1) &= \sum_{p=1}^2 \sum_{i=1}^2 \sum_{j=1}^2 |A_p|^2 A_i^* A_j \phi_{2cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[4a_{p2}\tau - (a_{i1} - a_{j1}) - 2(a_{i2} + a_{j2})\tau_1 \\
 &\quad - 3(a_{i3} - a_{j3})\tau_1^2]t + [6a_{p3}\tau - (a_{i2} - a_{j2}) \\
 &\quad - 3(a_{i3} + a_{j3})\tau_1]t^2 - (a_{i3} - a_{j3})t^3]\}\}.
 \end{aligned} \quad (B.1)$$

Applying FrFT to Eq. (B.1) along t axis, we can obtain $FrSCF_{2cross}(u, \tau, \tau_1)$. There are several cases needed to discuss in detail.

Case 2a. If $a_{i3} \neq a_{j3}$, Eq. (B.1) contains the third order phases and the energy of $FrSCF_{2cross}(u, \tau, \tau_1)$ can not be gathered to peak by product in FrFD. Therefore, the spurious peaks do not appear in this case.

Case 2b. If $a_{i3} = a_{j3}$ and $a_{p3} \neq a_{i3}$, Eq. (B.1) can be rewritten as

$$\begin{aligned}
 SCF_{2cross}(t, \tau, \tau_1) &= \sum_{p=1}^2 \sum_{i=1}^2 \sum_{j=1}^2 |A_p|^2 A_i^* A_j \phi_{2cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[4a_{p2}\tau - (a_{i1} - a_{j1}) - 2(a_{i2} + a_{j2})\tau_1]t \\
 &\quad + [6a_{p3}\tau - (a_{i2} - a_{j2}) - 6a_{i3}\tau_1]t^2]\}\}.
 \end{aligned} \quad (B.2)$$

Applying FrFT to Eq. (B.2) along t axis, we can obtain $FrSCF_{2cross}(u, \tau, \tau_1)$ and the spurious peak positions about α and u are as follows

$$\begin{cases} \cot \alpha = -2[6a_{p3}\tau - (a_{i2} - a_{j2}) - 6a_{i3}\tau_1] \\ u \csc \alpha = 4a_{p2}\tau - (a_{i1} - a_{j1}) - 2(a_{i2} + a_{j2})\tau_1 \end{cases} \quad (B.3)$$

We can find that the position of peak of cross term depends on τ from Eq. (B.3), of which $\cot \alpha$ is related with τ . It implies that the energy can not be gathered to be a spurious peak by the product with different τ .

Case 2c. If $a_{i3} = a_{j3}$ and $a_{p3} = a_{i3}$, Eq. (B.1) can be rewritten as

$$\begin{aligned}
 SCF_{2cross}(t, \tau, \tau_1) &= \sum_{p=1}^2 \sum_{i=1}^2 \sum_{j=1}^2 |A_p|^2 A_i^* A_j \phi_{2cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[-(a_{i1} - a_{j1}) + 4a_{p2}\tau - 2(a_{i2} + a_{j2})\tau_1]t \\
 &\quad + [-(a_{i2} - a_{j2}) - 6a_{i3}\tau_0]t^2]\}\}.
 \end{aligned} \quad (B.4)$$

Applying FrFT to Eq. (B.4) along t axis, we can obtain $FrSCF_{2cross}(u, \tau, \tau_1)$ and the spurious peak positions about α and u are as follows

$$\begin{cases} \cot \alpha = 2[(a_{i2} - a_{j2}) + 6a_{i3}\tau_0] \\ u \csc \alpha = -(a_{i1} - a_{j1}) + 4a_{p2}\tau - 2(a_{i2} + a_{j2})\tau_1 \end{cases} \quad (B.5)$$

From Eq. (B.5), the spurious peak position is always connected with τ closely whether $a_{i2} = a_{j2}$, $a_{i1} = a_{j1}$ or not. As a consequence, this cross term can be suppressed by our method.

The third category of cross term is

$$\begin{aligned}
 SCF_{3cross}(t, \tau, \tau_1) &= \sum_{i=1}^2 \sum_{j=1}^2 \sum_{q=1}^2 A_i^* A_j |A_q|^2 \phi_{3cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[-4a_{q2}\tau_1 + (a_{i1} - a_{j1}) + 2(a_{i2} + a_{j2})\tau \\
 &\quad + 3(a_{i3} - a_{j3})\tau^2]t + [-6a_{q3}\tau_1 + (a_{i2} - a_{j2}) \\
 &\quad + 3(a_{i3} + a_{j3})\tau]t^2 + (a_{i3} - a_{j3})t^3]\}\}.
 \end{aligned} \quad (B.6)$$

Because this category is similar to the second category, thus the details is omitted here.

The fourth category of cross term is

$$\begin{aligned}
 SCF_{4cross}(t, \tau, \tau_1) &= \sum_{i=1}^2 \sum_{j=1}^2 \sum_{p=1}^2 \sum_{q=1}^2 A_i A_j^* A_p^* A_q \phi_{4cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[(a_{i1} - a_{j1}) - (a_{p1} - a_{q1}) + 2(a_{i2} + a_{j2})\tau \\
 &\quad - 2(a_{p2} + a_{q2})\tau_1 + 3(a_{i3} - a_{j3})\tau^2]t \\
 &\quad + [(a_{i2} - a_{j2}) - (a_{p2} - a_{q2}) + 3(a_{i3} + a_{j3})\tau \\
 &\quad - 3(a_{p3} + a_{q3})\tau_1]t^2 + [(a_{i3} - a_{j3}) - (a_{p3} - a_{q3})]t^3]\}\}.
 \end{aligned} \quad (B.7)$$

According to Eq. (B.7), when compared with above three categories, it is the most complex cross term to be analyzed.

Case 4a. If $(a_{i3} - a_{j3}) - (a_{p3} - a_{q3}) \neq 0$, the FrFT is unable to gather the energy of signal to be peak due to the existence of the third phase t^3 in Eq. (B.7). So this case is similar to Case 2a.

Case 4b. If $(a_{i3} - a_{j3}) - (a_{p3} - a_{q3}) = 0$ and $a_{i3} \neq a_{j3}$, $a_{p3} \neq a_{q3}$, Eq. (B.7) can be rewritten as

$$\begin{aligned}
 SCF_{4cross}(t, \tau, \tau_1) &= \sum_{i=1}^2 \sum_{j=1}^2 \sum_{p=1}^2 \sum_{q=1}^2 A_i A_j^* A_p^* A_q \phi_{4cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[(a_{i1} - a_{j1}) - (a_{p1} - a_{q1}) + 2(a_{i2} + a_{j2})\tau \\
 &\quad - 2(a_{p2} + a_{q2})\tau_1 + 3(a_{i3} - a_{j3})\tau^2]t + [(a_{i2} - a_{j2}) \\
 &\quad - (a_{p2} - a_{q2}) + 3(a_{i3} + a_{j3})\tau - 3(a_{p3} + a_{q3})\tau_1]t^2]\}\}.
 \end{aligned} \quad (B.8)$$

Applying FrFT to Eq. (B.8), it is easy to obtain $FrSCF_{4cross}$ and the spurious peak positions are

$$\begin{cases} \cot \alpha = -2[(a_{i2} - a_{j2}) - (a_{p2} - a_{q2}) + 3(a_{i3} + a_{j3})\tau \\ \quad - 3(a_{p3} + a_{q3})\tau_1] \\ u \csc \alpha = (a_{i1} - a_{j1}) - (a_{p1} - a_{q1}) + 2[(a_{i2} + a_{j2})\tau \\ \quad - (a_{p2} + a_{q2})\tau_1] + 3(a_{i3} - a_{j3})\tau^2 \end{cases} \quad (B.9)$$

The spurious peak positions are related to τ and τ_1 , the PFrSCF method is capable to eliminate the spurious peaks luckily.

Case 4c. If $(a_{i3} - a_{j3}) - (a_{p3} - a_{q3}) = 0$ and $a_{i3} = a_{j3}$, $a_{p3} = a_{q3}$, Eq. (B.7) can be rewritten as

$$\begin{aligned}
 SCF_{4cross}(t, \tau, \tau_1) &= \sum_{i=1}^2 \sum_{j=1}^2 \sum_{p=1}^2 \sum_{q=1}^2 A_i A_j^* A_p^* A_q \phi_{4cross}(\tau, \tau_1) \\
 &\quad \cdot \exp\{j2\pi\{[(a_{i1} - a_{j1}) - (a_{p1} - a_{q1}) + 2(a_{i2} + a_{j2})\tau \\
 &\quad - 2(a_{p2} + a_{q2})\tau_1 + 3(a_{i3} - a_{j3})\tau^2]t + [(a_{i2} - a_{j2}) \\
 &\quad - (a_{p2} - a_{q2}) + 3(a_{i3} + a_{j3})\tau - 3(a_{p3} + a_{q3})\tau_1]t^2]\}\}.
 \end{aligned}$$

$$\begin{aligned} & \cdot \exp\{j2\pi[(a_{i1} - a_{j1}) - (a_{p1} - a_{q1}) + 2(a_{i2} + a_{j2})\tau \\ & - 2(a_{p2} + a_{q2})\tau_1]t + [(a_{i2} - a_{j2}) - (a_{p2} - a_{q2}) \\ & + 6a_{i3}\tau - 6a_{p3}\tau_1]t^2\} \end{aligned} \quad (\text{B.10})$$

Applying FrFT to Eq. (B.10), it is easy to obtain $\text{FrSCF}_{4\text{cross}}$ and the spurious peak positions

$$\begin{cases} \cot\alpha = -2[(a_{i2} - a_{j2}) - (a_{p2} - a_{q2}) + 6a_{i3}\tau - 6a_{p3}\tau_1] \\ u \csc\alpha = (a_{i1} - a_{j1}) - (a_{p1} - a_{q1}) + 2(a_{i2} + a_{j2})\tau \\ \quad - 2(a_{p2} + a_{q2})\tau_1. \end{cases} \quad (\text{B.11})$$

In terms of QFM signal, we have the coefficient $a_{i3} \neq 0$ and $a_{p3} \neq 0$. By analogy, the spurious peak positions are still related to τ and τ_1 from Eq. (B.11), the PFrSCF method is capable to eliminate the spurious peaks for this case.

Through the analysis of cross-term above all, it is easy to find that our PFrSCF method is always appropriate to suppress cross term whether term t^3 exist or not, which further demonstrates the effectiveness of the proposed method theoretically.

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